Lab Report Task 3:

To write this report, you can use proteus or hardware instruments taken from the lab (remember, if you take hardware instruments from the lab, then return it on the same day, otherwise the lab won't give you the instruments, may blacklist you. Room no. 507 is free for students to conduct experiments). If you use proteus, take screenshots of the circuits you prepared in the proteus and use it in the lab report. If you use hardwares from the lab, then take a clear picture of the overall circuit and then use it in your lab report.
Finally, you must submit the codes of each problem in the lab report. Check Lab reports samples/guidelines document to properly write your lab report. Maintain the formatting of the lab report as mentioned in the documents.

***** You have already been assigned a Raspberry Pi with an ID number. You must take that assigned Raspberry Pi while conducting your lab report tasks in 507 no. Room.**

The tasks for lab report 3: Normal Structure of the lab report (same for all the lab reports):

SL No.	Sections	What to Write
1.	Cover Page	Add a cover page and use the format uploaded in the drive (Lab Report Format (with cover page).docx)
2.	Objective	Write 2-3 lines using points to mention what was the main learning outcome of the report
3.	Components	Write what types of components you used.
4.	Theory	Write some theoretical background if needed for the problems you are going to solve in this report
5.	Problems mention in the tasks	Solve all the problems, attach circuit diagrams (take screenshots or using camera phone), attach codes in detail. Use comments in the code so that we can understand what did you write
6.	Discussion	Write 5-7 lines on what you learn and what problems you faced when solving these problems.

Provide the relevant circuit connections and their codes for all the problems mentioned below:

1) Connect a gas sensor to the Raspberry Pi and answer the followings:
    a) Write a python code to determine the gas volume in ppm data.
    b) Now, write a python code which will store that data in an excel file.

2) Add an Ultrasonic Sensor to your RPi and a LED. Write a python code that will determine the distance of any object from the Ultrasonic sensor and print it in the terminal. Also, if the object is very close to the sensor, the LED should light up.